University of Toronto

Auxiliary Representation Losses for Distractor-Robust Visual RL

Extending DrQ-v2 with Lightweight Self-Supervised Objectives

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10 min presentation + 5 min Q&A

Background: Learning RL from Pixels

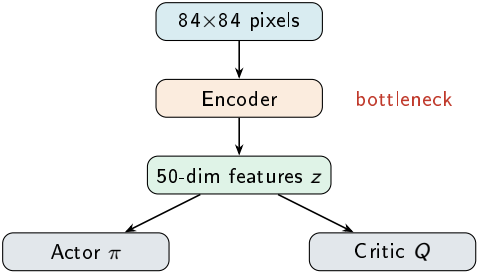
The challenge:
Visual RL gets **raw 84×84 pixels** (21,168 dims), not clean state vectors.

Solution: learn an encoder

- Pixels → 50-dim features. **Encoder quality = agent quality**
- Actor & Critic both depend on this representation

How to train it?

- DrQ-v2: only through critic loss
- Our idea: add **auxiliary losses** as extra signals



If the encoder fails to extract task-relevant features, both actor and critic will fail.

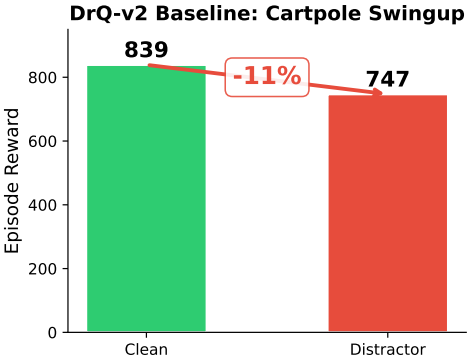
DrQ-v2: Strong but Fragile

DrQ-v2 (Yarats et al., ICLR 2022):

- **DDPG** + random-shift augmentation (± 4 px)
- Encoder learns to ignore small camera movements
- No auxiliary losses, no momentum encoders
- **SOTA on clean DMC tasks**

The Problem

Under **video distractors**, cartpole drops **$\sim$11%** (839 $\rightarrow$ 747). Random-shift cannot teach "ignore moving backgrounds".



Clean (blue) vs. Distractor (red)
Cartpole: 839 $\rightarrow$ 747 (-11%)

Research Question

Can lightweight auxiliary losses improve DrQ-v2's distractor robustness without changing its architecture?

Our Approach

Two auxiliary losses from our Assignment 1 paper reviews, added to the critic:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{critic}}}_{\text{original DrQ-v2}} + \alpha \cdot \underbrace{\mathcal{L}_{\text{aux}}}_{\text{our addition}}$$

Mod A — Task-agnostic

"Same state, different views
→ same encoding"

Inspired by SimSiam (self-supervised)

vs.

Mod B — Task-aware

"Similar Q-values
→ similar encoding"

Inspired by bisimulation metrics

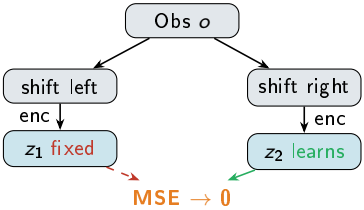
Mod A: Consistency Regularizer — Intuition

"If two views show the same physical state, their encodings must match."

Step by step:

1. Take one observation from the replay buffer
2. Apply two *different* random shifts (± 4 px)
3. Encode both shifted versions: get z_1 and z_2
4. Loss = MSE(z_1, z_2): **force them to match**

Why does this help with distractors?
The encoder learns: "if the physics didn't change, my output shouldn't change". This generalizes beyond pixel shifts — it encourages ignoring *any* visual variation that doesn't affect the task.



Same physical state
$\Rightarrow$ same encoding

Mod A: Technical Details

$$\mathcal{L}_{\text{consistency}} = \mathbb{E}_{o\sim\mathcal{B}}\left[\|\text{sg}[\text{enc}(f_1(o))] - \text{enc}(f_2(o))\|_2^2\right]$$

Three Practical Advantages

1. **Zero overhead**: DrQ-v2 already computes z_1, z_2 for actor/critic. We just add one MSE on top.
2. **No extra network**: just one loss term added to the critic.
3. **One hyperparameter**: $\alpha = 0.1$. Robust across $[0.01, 1.0]$ ($100\times$ range, only 4% performance variation).

Why stop-gradient (sg)?

Without it, the encoder would **cheat**: set all outputs to zero $\rightarrow z_1 = z_2 = 0$. Loss $= 0$, but representation is **useless**. This is called **representation collapse**.

Fix: freeze z_1 (stop-gradient). z_2 must match a *meaningful* target, not just collapse to zero.

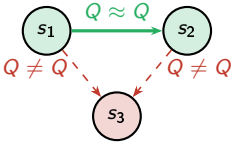
Mod B: Task-Conditional Contrastive Loss — Intuition

"States that play a similar role in the task should have similar encodings."

How it defines "similar role":

- Use **Q-values** as a proxy for behavioral similarity
- If $|Q_i - Q_j| < \epsilon$: these states are **positive pairs** (pull their encodings together)
- If $|Q_i - Q_j|$ is large: these are **negative pairs** (push their encodings apart)

vs. CURL (also contrastive):
CURL defines positives by *visual* augmentation.
Mod B defines positives by *task behavior* (Q-values).
$\Rightarrow$ Mod B is **task-aware**, not just vision-aware.



green = pull, red = push

This is a lightweight proxy for **bisimulation metrics**: states that "behave the same" should "look the same" to the encoder.

Mod B: Technical Details & Risk

$$\mathcal{L}_{\text{task}} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (i,j) \text{ pos. if } |Q_i - Q_j| < \epsilon$$

Problem 1: Computational Cost

Comparing every pair in a batch of 256:
$O(B^2) = 256^2 = 65{,}536$ comparisons.
Result: **+25% training time** per step.
(vs. Mod A: **0% overhead**)

Problem 2: Circular Dependency

1. Early Q-values are **noisy/wrong**
2. Wrong Q $\rightarrow$ wrong positive pairs
3. Wrong pairs $\rightarrow$ bad encoder training
4. Bad encoder $\rightarrow$ worse Q-values
5. $\rightarrow$ **vicious cycle, collapse**

Think of it as a **chicken-and-egg problem**.

Two competing ideas: Mod A is **simple & reliable**. Mod B is **smarter but fragile**.
Kean will now show you which one wins.

Experimental Setup

DMC-GB Distractors: background pixels replaced by video frames. Same physics & reward.

Task	Frames	Seeds
`cartpole_swingup`	500K	3
`walker_walk`	1M	1
`cheetah_run`	1M	1

$\times$ 2 settings $\times$ 4 methods = **24 runs**

3 seeds for cartpole; 1 seed for walker/cheetah (8–10h each)

4 Methods:

1. Baseline (DrQ-v2)
2. + Mod A ($\alpha = 0.1$)
3. + Mod B ($\alpha = 0.1$)
4. + Mod A+B (both)

Overhead per step:

Baseline: 48ms | Mod A: 46ms (**0%**)

Mod B: 60ms (**+25%**) | A+B: 95ms (**+97%**)

Cartpole Swingup — 3-Seed Results

Method	Clean	Distractor	Retention
DrQ-v2 (baseline)	839.0 ± 26.8	746.9 ± 39.0	89.0%
SuccessGreen!20 + Mod A	829.3 ± 29.2	782.6 ± 7.8	94.4%
+ Mod B†	575.2 ± 354.5	751.5 ± 29.6	—
+ Mod A+B	813.8 ± 45.5	764.5 ± 30.3	93.9%

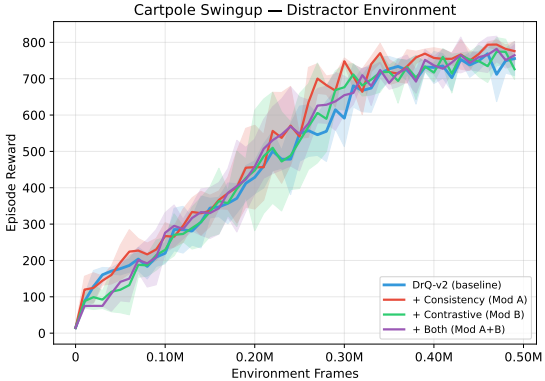
†Seed 2 collapse (reward 74.8); excluding: 825.4 ± 38.9

Mod A
Best distractor reward
Smallest variance ($\pm$7.8)

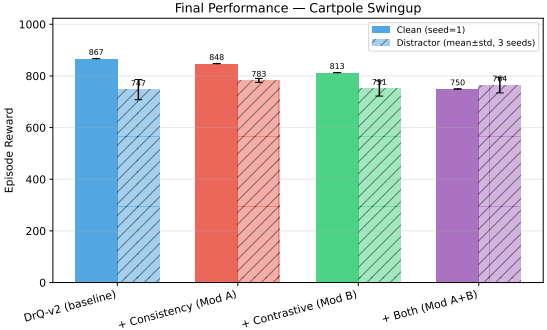
Mod B
1/3 seeds collapsed
Extreme $\pm$354.5 variance

Effect size
Cohen's $d = 1.27$
(Mod A vs. baseline)

Learning Curves & Final Performance



Distractor setting — Mod A converges fastest with tightest confidence bands



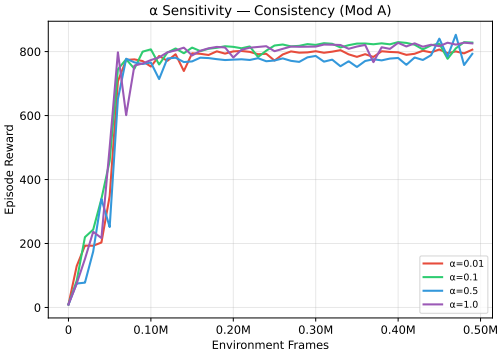
Solid = clean, hatched = distractor (cartpole: 3-seed mean)

Cross-Task Generalization (Seed=1)

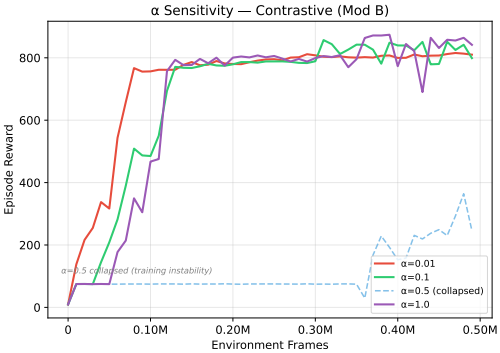
	Cartpole (500K)		**Walker** (1M)		**Cheetah** (1M)	
Method	Clean	Dist.	Clean	Dist.	Clean	Dist.
Baseline	867	702	958	959	743	596
+ **Mod A**	848	746	968	947	657	410
+ Mod B	853	731	964	**27**	773	626
+ Mod A+B	750	770	907	951	705	618

- **Mod A**: best walker distractor, but **hurts cheetah** (409.5 vs. 595.6)
- **Mod B**: walker distractor collapse (26.6), best clean cheetah (773.2)
- **No single auxiliary loss dominates all tasks**

α Sensitivity Ablation (Cartpole)

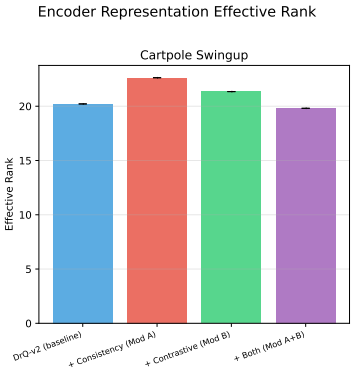


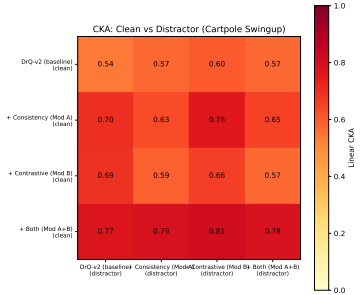
Mod A: Robust — all $\alpha \in [0.01, 1.0]$ converge to ∼800–830



Mod B: Fragile — α=0.5 collapses to ∼277; non-monotonic failure

Representation Analysis: Rank & Invariance





Effective Rank (SVD entropy)

A: **22.5** > B: 21.2 > Base: 20.2 > A+B: 19.8

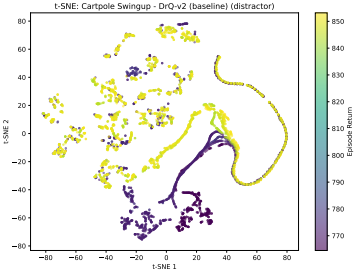
Higher rank = more expressive encoder

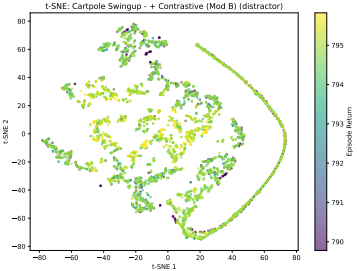
CKA (clean–distractor similarity)

A+B: **0.78** > B: 0.66 > A: 0.63 > Base: 0.54

Higher CKA = encoder ignores distractors

Representation Analysis: Latent Space (t-SNE)





Baseline (distractor)

Wide return spread (770–855)

Mod B (distractor)

Tight return clustering (790–796)

Paradox: Mod B produces the best representations, yet is the most unstable method

Why Mod A Works & Mod B Fails

Mod A — Recommended

- **Zero overhead**: one MSE term
- Highest effective rank (22.5)
- Robust to α: 0.01–1.0
- Smallest variance ($\pm$7.8), 94.4% retention

Caveat: hurts cheetah (409.5 vs. 595.6)

Mod B — Unstable

Circular dependency:
good pairs $\rightarrow$ good Q $\rightarrow$ good encoder $\rightarrow$ pairs
Failures: 1/3 seeds collapsed; α=0.5 fails; walker: **26.6**

Paradox: A+B has highest CKA (0.78) but *not* highest reward $\Rightarrow$ **invariance** $\neq$ **performance**

Conclusion & Future Work

Key Findings

1. DrQ-v2 drops ∼11% under video distractors
2. **Mod A**: best practical choice — zero overhead, robust α
3. **Mod B**: best representations but unstable
4. No universal auxiliary loss — task-dependent
5. Invariance $\neq$ performance

Future Work

- Adaptive α via Q-confidence
- Warm-start Mod B after Q converges
- High-dimensional manipulation tasks
- Real-world distractors (Kinetics)

Thank you for your attention

Bangcheng Wang	Motivation, Method, Setup
Kean Chen	Results, Analysis, Conclusion

Questions?